An Entanglement Mechanism for the Mach–Sciama Relation: Newton's Constant from Cosmic Boundary Conditions

Abstract

The relation $G \sim c^2R/M$ between Newton's gravitational constant and cosmic boundary conditions is well-known in the Mach–Sciama tradition: in any flat critical-density cosmology, $Gc^2 = (\text{const}) \cdot c^4R/M$ follows from the Friedmann equation. What is *not* explained in standard cosmology is *why* the universe should sit at this self-consistency point. We propose a quantum entanglement mechanism — orthogonal jerk creating helical trajectories, with bond energy $\hbar c/r$ and torque geometry from the Bisognano–Wichmann theorem — that predicts the specific prefactor $\gamma = 4/5$ from the modular Hamiltonian of the cosmic reduced density matrix. The relation

$$G \;=\; \frac{4}{5}\,\frac{c^2 R_{\text{universe}}}{M_{\text{universe}}} \tag{1}$$

gives a value consistent with the measured G to within current cosmic parameter uncertainties.

Scope. This is not a free derivation of G. The cosmic inputs R_{universe} and M_{universe} are themselves determined within general-relativistic cosmology and contain G implicitly (Section 2). The substantive content is the prediction of the prefactor $4/5$ from the modular Hamiltonian, and the mechanism itself, not the numerical value of G. A cleaner G-independent test of the same mechanism, the exact derivation of the Bohr magneton at the electron scale, is given in a companion paper.

1 THEORETICAL FOUNDATION

The derivation rests on five independent results from quantum field theory and conformal field theory.

1.1 Bisognano–Wichmann Theorem (1975/1976)

The modular Hamiltonian for a half-space in relativistic QFT is the generator of Lorentz boosts perpendicular to the entangling surface:

$$K = 2\pi \int_{x>0} \mathrm{d}^{d-1}x \; x \; T_{00}(x) \tag{2}$$

This is a **theory-independent** result. It applies to any relativistic QFT, interacting or free. The modular Hamiltonian is always a weighted integral of the stress-energy tensor T_{00}, with the weight being the distance x from the entangling surface.

Physical meaning: Entanglement creates a boost-rotation geometry, not a time-translation. The nucleon's spin faces orthogonally to the displacement toward the partner.

1.2 Casini–Huerta–Myers (2011)

For a spherical region of radius R in a d-dimensional CFT, the modular Hamiltonian is:

$$K = 2\pi \int_{|\vec{x}|<R} \mathrm{d}^{d-1}x \; \beta(r) \; T_{00}(\vec{x}) \tag{3}$$

where the weight function is:

$$\beta(r) = \frac{R^2 - r^2}{2R} \tag{4}$$

This is a parabola that vanishes at the boundary $r = R$ and is maximal at the center $r = 0$. The result is derived by conformally mapping the ball to a half-space and applying the BW theorem.

1.3 Cardy–Tonni (2016)

For an interval of length l in 1+1D CFT, the entanglement Hamiltonian is derived via conformal mapping from a punctured strip to an annulus:

$$H_E = 2\pi \int_0^l \mathrm{d}x \; \beta(x) \; T_{00}(x) \tag{5}$$

where the inverse temperature profile is:

$$\beta(x) = \frac{l}{\pi} \frac{\cos(\pi x_0/l) - \cos(\pi x/l)}{\sin(\pi x_0/l)} \tag{6}$$

For the full interval ($x_0 = l/2$), $\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$, so:

$$\beta(x) = -\frac{l}{\pi} \cos(\pi x/l) \tag{7}$$

The energy scale is set by the inverse of the interval length: $E \sim \hbar c/l$.

1.4 Bernard et al. (2024)

The modular Hamiltonian for inhomogeneous free-fermion chains has a commuting operator T_A whose eigenvalues approximate the entanglement spectrum:

$$\epsilon_k = \frac{2\pi\lambda_k}{\text{scale} \cdot N^2} \tag{8}$$

For the Racah chain (most general case), the full inverse temperature $\beta(x) \propto x(L - x)$ is parabolic. When the filling fraction is low ($\rho \ll 1/2$), the Fermi velocity vanishes outside a narrow active region $[x_-, L]$ near the boundary. Setting $u = (L - x)/L \in [0, 1 - x_-/L]$ and re-expressing β on the active interval, the dominant term is $\beta(r) \propto (1 - r/R)^2$ to leading order in the ratio $(L - x_-)/L$ [1]:

$$\beta(r) = (1 - r/R)^2 \tag{9}$$

1.5 Huerta & van der Velde (2023–2024)

The modular Hamiltonian for a spherical region in d-dimensional CFT can be obtained by dimensional reduction to the radial semi-infinite line. When a d-dimensional free massless field is decomposed into angular modes, the modular Hamiltonian decomposes as:

$$K = \sum_{\ell,\vec{m}} K_{\ell,\vec{m}} \tag{10}$$

where each mode contribution $K_{\ell,\vec{m}}$ is **exactly** the dimensional reduction of the parent CFT modular Hamiltonian. The weight function $\beta(r)$ is **preserved** under dimensional reduction.

Critically, the 1D reduced theories are **non-conformal** (they carry a $1/r^2$ potential from the angular Laplacian), yet their modular Hamiltonian remains local in the energy density with the **same weight function** as the d-dimensional CFT. This is proven for both scalar fields and Dirac fermions [3, 2].

2 CIRCULARITY AND SCOPE

Before presenting the mechanism it is essential to be explicit about what this calculation does and does not establish.

2.1 The Mach–Sciama Relation Is Not Novel

In a spatially flat universe at critical density, the Friedmann equation gives $\rho_c = 3H^2/(8\pi G)$. The total mass-energy in the Hubble volume is then

$$M_H \;=\; \rho_c \cdot \frac{4}{3}\pi R_H^3 \;=\; \frac{c^2 R_H}{2G}, \qquad R_H = c/H, \tag{11}$$

which immediately rearranges to $G = c^2 R_H/(2M_H)$. A relation of the form $G \sim c^2 R/M$ is therefore present in any standard cosmology; it has been discussed since Sciama (1953) and is the structural premise underlying Brans–Dicke theory and several modern emergent-gravity programs. *Reproducing this structural relation is not, by itself, new physics.*

2.2 The Inputs Are Not G-Independent

In standard cosmology, both R_{universe} and M_{universe} are determined within the GR/ΛCDM framework and depend on G:

- R_{universe} is the proper distance to the particle horizon, computed from a Friedmann integral that contains G through the matter density. Stellar age estimates (white dwarf cooling, globular cluster main-sequence turn-off) provide a partially G-independent age constraint, but the radius $R_{\text{universe}} = c \cdot t_{\text{age}}$ is at best a kinematic proxy.
- M_{universe} in ΛCDM is set by the closure condition $\Omega_{\text{tot}} = 1$, which contains G explicitly. Dark matter mass is determined by gravitational dynamics, so it cannot be measured independently of G. Only the baryonic component ($\Omega_b \approx 0.049$, from BBN and the CMB) is constrained by non-gravitational physics.

The numerical match between $\frac{4}{5}c^2 R_{\text{universe}}/M_{\text{universe}}$ and the measured G is therefore *not* evidence that the entanglement mechanism predicts the correct value of G from G-free inputs. It is evidence that the entanglement mechanism is consistent with the Mach–Sciama relation to within the prefactor.

2.3 Alternate Choices of R_{universe} and M_{universe}

One might hope to escape the circularity by picking inputs that are less G-dependent. None of the obvious choices succeed:

Table 1: Alternate choices of R and their G-independence.

Choice for R	G-independent?	What the claim becomes
Particle horizon (∼46 Gly)	No — Friedmann integral with G inside	Equivalent to the present paper
Hubble radius c/H_0	Partially — H_0 from redshift-distance is the cleanest	Equivalent to "universe is at critical density"
CMB last-scattering radius	No — requires GR perturbation theory to interpret	Buys nothing
Local causal diamond	Yes in principle	Promising; addressed in Sec. 2.4

Table 2: Alternate choices of M and their G-independence.

Choice for M	G-independent?	What the claim becomes
Baryonic mass only	Partially — BBN + atomic spectra are largely G-free	Predicts wrong G by factor of ~ 20 ($\Omega_b \approx 0.049$)
Baryon + dark matter	No — DM is gravitationally inferred	Reduces error but stays circular
Total energy budget	No — $\rho_c = 3H^2/(8\pi G)$ is explicitly circular	Identity, not derivation
Komar/ADM/Misner-Sharp	No — GR mass notions	Inside what we'd derive

The most revealing entry is "baryonic mass only." If the relation $G = (4/5)\, c^2 R_{\text{universe}}/M_{\text{universe}}$ were a genuine prediction from non-gravitational inputs, the value of M_{universe} that "works" should be the one that comes from non-gravitational measurements (BBN, atomic spectra, stellar populations). It isn't. The value that makes the relation match G requires including dark matter and dark energy, both of which are gravitationally inferred. The match is therefore even more circular than the headline calculation suggests: most of the mass-energy "input" is itself a G-laden output.

2.4 The Local Test: Earth as a Causal Patch

The most decisive way to ask whether $G = \gamma\, c^2 R/M$ is a real prediction is to apply it to a system that is not the entire observable universe. If the relation is a local statement, it should hold for any causal patch. Try Earth:

$$M_E = 5.97 \times 10^{24}\ \text{kg}, \tag{12}$$

$$R_E = 6.37 \times 10^{6}\ \text{m}, \tag{13}$$

$$G_{\text{eff}}^{\text{Earth}} = \frac{4}{5}\frac{c^2 R_E}{M_E} \approx 7.7 \times 10^{25}\ \text{m}^3/(\text{kg}\,\text{s}^2). \tag{14}$$

This is off from the measured $G \approx 6.67 \times 10^{-11}$ by roughly **36 orders of magnitude**. The relation $G = \gamma\, c^2 R/M$ is therefore *not* a local prediction. Plug in any patch other than the whole universe, and it gives nonsense.

The reason: at the cosmic scale, $M_{\text{universe}}/R_{\text{universe}} \approx c^2/(2G)$ because the universe sits at critical density — the very condition that contains G. The relation works only at the one scale where the inputs are forced to satisfy the closure condition that contains G.

This is the strongest form of the circularity critique: the "derivation" produces the right answer at exactly one scale, the one defined by an equation that contains the constant being "derived."

2.5 What Is and Is Not Claimed

What is claimed:

1. There is a physically motivated entanglement mechanism (orthogonal jerk + helical trajectory + bond energy $\hbar c/r$) whose application at the electron Compton scale exactly reproduces the Bohr magneton, with no G as input. This is the only clean numerical test of the mechanism (companion paper).

2. The same mechanism, when applied to the cosmic ensemble with the standard ΛCDM inputs, reproduces the Mach–Sciama relation $G \sim c^2 R/M$ with a specific predicted prefactor $\gamma = 4/5$ derived from the modular Hamiltonian of free fermions on a Racah chain at low filling fraction.

What is not claimed:

1. This is *not* a derivation of G from G-free inputs. The cosmic parameters used are not G-independent, and the relation evaluated on any local patch (e.g. Earth) gives nonsense by ~ 36 orders of magnitude.

2. The 0.21% point-estimate match between $(4/5)c^2 R_{\text{universe}}/M_{\text{universe}}$ and the measured G is therefore a consistency check at the one scale where the inputs are forced by the critical-density condition, not a precision prediction.

3. The mechanism does not, on its own, replace general relativity, and the present treatment does not constitute a local field-theoretic derivation of Newtonian gravity from entanglement.

The cleanest empirical leg of the program is therefore not this gravity paper, but the Bohr magneton derivation at the electron scale, which is local, uses no G, and matches an experimentally measured quantity to high precision. The reader should weight the program accordingly.

3 THE BOND ENERGY $E_{\text{BOND}} = \hbar C/R$

3.1 Dimensional Analysis

The modular Hamiltonian K is dimensionless (it appears in $\rho_A = e^{-K}/Z$). The weight β has dimensions $[L]$ (boost parameter = proper distance). The energy density T_{00} has dimensions $[E]/[L]^d$. The volume element $\mathrm{d}^d x$ has dimensions $[L]^d$.

$$K = 2\pi \int \mathrm{d}^d x \; \beta(x) \; T_{00}(x) \tag{15}$$

The integral has dimensions $[L]^d \cdot [L] \cdot [E]/[L]^d = [E] \cdot [L]$. For K to be dimensionless:

$$\frac{[E] \cdot [L]}{\hbar c} = \text{dimensionless} \implies E \sim \frac{\hbar c}{L} \tag{16}$$

This is the universal scaling: the entanglement energy between two subsystems at separation L is $\hbar c/L$, up to a dimensionless coefficient.

3.2 Why Free Fermions Apply to Nucleons

The Bernard et al. calculation works with free fermions. Nucleons interact via the strong force. Why does the free fermion result apply?

1. **BW theorem is theory-independent:** The modular Hamiltonian structure $K = 2\pi \int \beta T_{00}$ applies to any relativistic QFT, interacting or free. The specific form of T_{00} does not change the local structure.

2. **Dimensional analysis is theory-independent:** The scaling $E \sim \hbar c/L$ depends only on the dimensions of T_{00} and β, which are universal.

3. **Bernard et al. is a verification, not an assumption:** It confirms the universal structure in a concrete solvable model.

4. **EFT argument:** At cosmic scales, only massless degrees of freedom matter. Detailed QCD physics is integrated out and appears only in the normalization, not in the $1/r$ scaling.

5. **Cross-check:** At the nucleon scale ($r \sim 1$ fm), $\hbar c/r \sim 200$ MeV, matching Λ_{QCD}. This agreement at both nucleon and cosmic scales is independent confirmation.

3.3 Numerical Verification

Table 3: Bond energy $E_{\mathrm{bond}} = \hbar c/r$ across scales.

Scale	r	$E_{\mathrm{bond}} = \hbar c/r$	Physical interpretation
Nucleon	1 fm	3.2×10^{-11} J (0.2 MeV)	Strong force scale (Λ_{QCD})
Earth	6371 km	5.0×10^{-33} J	3.1×10^{-14} eV
Cosmic	4.4×10^{26} m	7.2×10^{-53} J	Gravitational binding scale

4 THE GEOMETRIC FACTOR $\gamma = 4/5$

4.1 The 1D Racah Chain Result

The cosmic filling fraction is $\rho = \Omega_b = 0.049$ (the baryon fraction). In the Racah chain framework, this maps to parameters:

$$x_1 = \frac{\Omega_{\mathrm{DM}}}{\Omega_b} = 5.47, \quad x_2 = \frac{\Omega_{\mathrm{DE}}}{\Omega_b} = 13.94, \quad x_3 = \frac{\Omega_{\mathrm{DM}}}{\Omega_{\mathrm{DE}}} = 0.39 \tag{17}$$

The Fermi velocity $v_F(x)$ is non-zero only on the active region $[x_-, x_+] = [0.868, 1.000]$, a narrow band near the boundary. The modular Hamiltonian weight on this region is quadratic:

$$\beta_{1\mathrm{D}}(x) = (1 - x/L)^2 \tag{18}$$

4.2 The 3+1D Extension

The Huerta & van der Velde theorem proves that the 1D modular Hamiltonian on the radial line is the dimensional reduction of the d-dimensional sphere. The weight function $\beta(r)$ is **preserved** under dimensional reduction.

In 3+1D, the spherical volume element $r^2 \mathrm{d}r$ modifies the weighted average. The weighted average of $1/r$ with the quadratic weight is:

$$\left\langle \frac{1}{r} \right\rangle_{3\mathrm{D}} = \frac{\int_0^R \beta(r) \cdot r \, \mathrm{d}r}{\int_0^R \beta(r) \cdot r^2 \, \mathrm{d}r} \tag{19}$$

For $\beta(r) = (1 - r/R)^2$:
Numerator:

$$\int_0^R (1 - r/R)^2 \cdot r \, \mathrm{d}r = R^2 \int_0^1 u^2(1-u) \, \mathrm{d}u = R^2 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{R^2}{12} \tag{20}$$

Denominator:

$$\int_0^R (1 - r/R)^2 \cdot r^2 \, \mathrm{d}r = R^3 \int_0^1 u^2(1-u)^2 \, \mathrm{d}u = R^3 \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right) = \frac{R^3}{30} \tag{21}$$

Result:

$$\left\langle \frac{1}{r} \right\rangle = \frac{R^2/12}{R^3/30} = \frac{5}{2R} \tag{22}$$

The geometric factor is:

$$\gamma = \frac{2}{\langle 1/r \rangle \cdot R} = \frac{2}{5/2} = \frac{4}{5} \tag{23}$$

4.3 Sensitivity to the Weight Exponent

The quadratic weight $a = 2$ is determined by the Racah chain physics (low filling fraction $\rho = 0.049$), not by fitting to G. A scan of nearby exponents shows the result is sensitive: the error grows rapidly as a deviates from 2.

Table 4: Sensitivity of γ to the weight exponent a in $\beta(r) = (1 - r/R)^a$.

a	$\langle 1/r \rangle \, R$	γ	Error vs measured G
1.8	2.400	0.8333	4.39%
1.9	2.450	0.8163	2.26%
2.0	**2.500**	**0.8000**	0.21%
2.1	2.550	0.7843	1.75%
2.2	2.600	0.7692	3.64%

The sensitivity confirms that the quadratic weight is not an arbitrary choice: a small deviation from $a = 2$ would produce a noticeably worse agreement with the measured value of G.

5 THE MECHANISM

5.1 Cosmic Entanglement Structure

In the cosmic reduced density matrix, every nucleon is entangled with every other nucleon. The angular momentum budget per nucleon is:

$$s = \frac{\sqrt{3}}{2}\hbar \tag{24}$$

This budget is distributed across N_{universe} partners, with the allocation following the modular Hamiltonian structure ($\hbar c/r$).

5.2 Torque Geometry

The BW theorem shows that the modular flow is a boost rotation, not a time translation. The nucleon's spin faces orthogonally to the displacement toward the partner. This is the "torque geometry" of entanglement.

5.3 Unresolved Entanglement

When the per-pair coupling between two particles is far below unity, the pair has not collapsed to a definite correlation. The entanglement amplitude still exists but remains unresolved. For gravity, this coupling is $\alpha_{\text{grav}} \approx 5.9 \times 10^{-39}$, so essentially all nucleon pairs are in this unresolved regime.

This is not zero force. The unresolved amplitude carries the cosmic entanglement field: each nucleon is entangled with N_{universe} other nucleons. The balance of angular momentum is the rest of the universe. In QFT language, this is the reduced density matrix obtained by tracing out the complement. The universe's isotropic entanglement field is the baseline; any local mass breaks the isotropy.

5.4 Gravity as Anisotropy

The boost rotation back toward the mass is the gravitational force. Gravity is not a fundamental force—it is the anisotropy in the cosmic entanglement field created by local mass concentrations.

6 THE DERIVATION

Terminology note. Throughout this section we use the symbol C to denote the per-pair entanglement coupling (the strength with which two nucleons share entanglement). This is *not* the Wootters concurrence of two-qubit quantum information theory. The two should not be conflated.

6.1 Setup

Each nucleon carries angular momentum $s = \sqrt{3}/2 \cdot \hbar$. From the modular Hamiltonian (Sections 1–2), the bond energy between two nucleons at separation r is:

$$E_{\text{bond}} = \kappa \frac{\hbar c}{r} \tag{25}$$

where κ is an order-unity geometric coefficient. The BW theorem sets the torque geometry: the nucleon faces orthogonally to the displacement toward the partner. The isotropic cosmic background cancels, leaving only the anisotropy from local mass concentrations.

6.2 Per-Pair Coupling from Energy Balance

The per-pair coupling C is determined by the total entanglement energy of the universe. The universe's mass-energy is stored as entanglement energy across all nucleon pairs:

$$E_{\text{entanglement}} = \frac{N_{\text{universe}}^2}{2} \cdot C \cdot \kappa\hbar c \cdot \left\langle \frac{1}{r} \right\rangle \tag{26}$$

where $\langle 1/r \rangle = 5/(2R_{\text{universe}})$ from the quadratic weight. Equating to the universe's mass-energy $M_{\text{universe}}c^2$:

$$\frac{N^2}{2} \cdot C \cdot \kappa\hbar c \cdot \frac{5}{2R} = Mc^2 \tag{27}$$

Solving for C and substituting $N = M/m_p$:

$$C = \frac{4}{5} \frac{m_p^2 c R_{\text{universe}}}{\kappa M_{\text{universe}} \hbar} \tag{28}$$

6.3 G Emerges

The force per nucleon pair is:

$$F_{\text{pair}} = C \cdot \kappa \frac{\hbar c}{r^2} \tag{29}$$

The total force between two masses M and m at distance r:

$$F = \frac{M}{m_p} \cdot \frac{m}{m_p} \cdot C \cdot \kappa \frac{\hbar c}{r^2} \tag{30}$$

Substituting eq. (28):

$$F = \frac{4}{5} \cdot M \cdot m \cdot \frac{c^2 R_{\text{universe}}}{M_{\text{universe}}} \cdot \frac{1}{r^2} \tag{31}$$

Comparing to Newton's law $F = GMm/r^2$:

$$G = \frac{4}{5} \frac{c^2 R_{\text{universe}}}{M_{\text{universe}}} \tag{32}$$

Note: $\hbar$, m_p, and κ all cancel. The final expression depends only on c, R_{universe}, and M_{universe}.

7 NUMERICAL CONSISTENCY CHECK

Using cosmic inputs:

$$c = 2.998 \times 10^8 \text{ m/s} \tag{33}$$
$$R_{\text{universe}} = 4.4 \times 10^{26} \text{ m} \tag{34}$$
$$M_{\text{universe}} = 4.73 \times 10^{53} \text{ kg} \tag{35}$$

the predicted relation gives

$$G_{\text{predicted}} = \frac{4}{5} \frac{(2.998 \times 10^8)^2 \cdot (4.4 \times 10^{26})}{4.73 \times 10^{53}} = 6.688 \times 10^{-11} \text{ m}^3/(\text{kg}{\cdot}\text{s}^2), \tag{36}$$

to be compared with $G_{\text{measured}} = 6.674 \times 10^{-11}$ m^3/(kg·s^2). The ratio is 1.0021.

This check should be read in the light of Section 2: the inputs R_{universe} and M_{universe} are not independent of G, and each carries roughly 5–10% uncertainty in standard cosmology. The headline 0.21% agreement is therefore a consistency check, not a precision result. The honest statement is: the predicted prefactor 4/5, combined with the standard cosmic parameters, is consistent with the measured value of G to well within those parameter uncertainties.

A genuine prefactor test would require an independent constraint on $M_{\text{universe}}/R_{\text{universe}}$ that does not pass through G. We are not aware of one that is currently tight enough to discriminate, say, 4/5 from 1/2 or 1 at the few-percent level.

8 STATUS OF THE PREDICTIONS

Honest accounting of what the mechanism currently predicts cleanly, predicts structurally, and does not yet predict.

8.1 Cleanly Predicted (G-Free): Bohr Magneton

Applied at the electron scale, the same helical-jerk mechanism predicts

$$\mu_B = \frac{e\hbar}{2m_e} \tag{37}$$

exactly, with no fitted parameters and using only e, $\hbar$, m_e, and c — none of which involve G. This is the cleanest test of the mechanism and is treated in detail in the companion paper.

8.2 Structurally Predicted (Quantitatively Off): Frame Dragging

The boost generator structure (BW theorem) implies that rotating masses drag the local entanglement field, producing a Lense–Thirring-like effect. The mechanism predicts the correct *structure*: frame dragging is a boost effect of the modular flow under rotation, not a separate phenomenon.

The *quantitative* match is currently off by roughly a factor of 10^3. A naive collective treatment (random walk of $\sqrt{N_{\text{Earth}}}$ nucleons) does not reproduce $\omega_{\text{FD}}^{\text{GR}} = 2GJ/(c^2R^3) \approx 1.9 \times 10^{-14}$ rad/s. Closing this gap requires a more precise treatment of the collective contribution at the nucleon level, and is open work.

8.3 Order-of-Magnitude: Dark Energy

The unresolved entanglement field at the cosmic scale gives an energy density of order $\rho_{\text{DE}} \approx 8.1 \times 10^{-11}$ J/m^3, compared to the observed $\rho_{\text{DE}}^{\text{obs}} \approx 6.9 \times 10^{-10}$ J/m^3. The prediction is within an order of magnitude. The factor of ~ 8 is consistent with the holographic picture (dark energy stored on the cosmic horizon, a 2D surface, rather than distributed through the bulk volume), but a precise holographic accounting is beyond the present scope.

8.4 Not a Free Prediction: α_{grav}

The standard gravitational coupling $\alpha_{\text{grav}} = Gm_p^2/(\hbar c) \approx 5.9 \times 10^{-39}$ follows from the predicted relation for G together with m_p, $\hbar$, and c. It is therefore consistent with the framework but not an independent prediction — it inherits the same circularity as G itself.

9 COMPARISON TO KOLMOGOROV'S FOUR-FIFTHS LAW

The factor 4/5 also appears as an exact coefficient in Kolmogorov's four-fifths law for fully developed turbulence, where the third-order longitudinal velocity structure function satisfies $\langle(\delta u_{\|})^3\rangle = -\frac{4}{5}\varepsilon r$ in the inertial range. This is one of the few exact results in turbulence theory, deriving from energy conservation in a cascade across scales. The recurrence of 4/5 in two distinct energy-transfer-across-scales problems (turbulent cascade in 3D, entanglement-energy distribution across the cosmic 3+1D modular Hamiltonian) is suggestive but not by itself a derivation; we note the parallel as a possible structural connection.

10 TOWARD A LOCAL DERIVATION

Section 2.4 showed that the cosmic relation $G = (4/5)\,c^2R/M$ is not local. The right next step, if the mechanism is to be more than a fit at one scale, is to attempt a local derivation in the spirit of the Jacobson–Padmanabhan–Verlinde program. We sketch what would be required and what is currently missing.

10.1 The Jacobson Program

(author?) [4] showed that Einstein's equation can be derived from the Clausius relation $\delta Q = T\,dS$ applied to local Rindler horizons, given the Bekenstein–Hawking entropy $S = A/(4\ell_P^2)$. **(author?)** [5] and **(author?)** [6] extended this to "entropic gravity" programs in which Newtonian gravity emerges from local entanglement entropy variation across a screen.

Crucially, in all of these programs G enters through the *coefficient* of the area law, $1/(4\ell_P^2)$. None of them *derives* the value of G; they derive Einstein's equation *given* the entropy–area coefficient. The Planck length itself contains G by definition ($\ell_P = \sqrt{\hbar G/c^3}$). So the deepest known programs of this type still require an external input that fixes the gravitational coupling.

10.2 What a Local Version of Our Mechanism Would Need

For our specific entanglement-torque mechanism to constitute a local derivation of G, three things would have to be true:

1. **A universal entanglement bond density.** There must exist a local quantity — call it η, with units of [energy] $\cdot$ [length]$^{-3}$ $\cdot$ [pair]$^{-1}$, or equivalent — that is the same in every vacuum patch and is determined by QFT alone.

2. **Newtonian acceleration from torque-of-jerk integrated over the patch.** The same orthogonal-jerk + helical-trajectory geometry, summed over the local entanglement structure, must produce $\mathbf{a} = -\nabla\Phi$ with $\nabla^2\Phi = 4\pi G\,\rho$ in the weak-field limit, with a specific dimensionless proportionality constant that does *not* depend on the patch's size or contents.

3. **η computed without G.** The crucial step. The entanglement bond density η must be derivable from purely QFT inputs (modular Hamiltonians, area-law coefficients) without using G as an input. If η requires the Planck length to define its cutoff, the program inherits the same circularity as Jacobson's.

We can offer (1) and (2) at the level of structural sketch. We do *not* have (3), and to our knowledge nobody else does either. This is the open problem that, if solved, would convert the present program from a consistency check into a derivation.

10.3 Why the Bohr Magneton Is the Real Local Test

Although a local derivation of G is an open problem, the same mechanism applied at the electron Compton scale produces an exact, local, G-independent result: $\mu_B = e\hbar/(2m_e)$ (companion paper). This is the real evidence that the mechanism is on solid local ground; what's missing is the bridge from "the mechanism is locally real" to "the mechanism, applied to the cosmic ensemble, fixes G."

The honest position is therefore: the local seed is proven (Bohr magneton). The orchard's shape (Mach–Sciama at cosmic scale) is consistent with the same seed and with a specific predicted prefactor, but does not by itself constitute a derivation of G.

11 COMPUTATIONAL VERIFICATION

All derivations are implemented in Python scripts:

- `derive_G.py`—Main G derivation
- `derive_bond_energy.py`—$E_{\text{bond}} = \hbar c/r$ derivation
- `derive_3d_extension.py`—3+1D extension via Huerta & van der Velde
- `constants.py`—Physical constants and helper functions

All scripts run cleanly and produce consistent results.

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